SUKUMAR BV

+91 9380964393 | sukumarbv27@gmail.com | SukumarBV | Portfolio | Github

SUMMARY

AI & Machine Learning Engineering student (CGPA: 8.88/10) with experience building and deploying end-to-end machine learning applications. Skilled in Python, FastAPI, PyTorch, TensorFlow, and Docker, with project experience spanning NLP, Computer Vision, MLOps, and predictive analytics. Passionate about turning data-driven ideas into practical solutions and eager to contribute to real-world AI products while continuing to grow as an engineer.

SKILLS

Languages: Python, Java, JavaScript, SQL, C++
AI / Machine Learning: NLP, Text-to-SQL, Computer Vision, LSTM, Scikit-learn, PyTorch, TensorFlow, CVZone
Generative AI: LLMs, Prompt Engineering, RAG, Vector Databases, Embeddings, Gemini API, Hugging Face, AI Agents
Frameworks & Libraries: FastAPI, React, SQLAlchemy, OpenCV, SciPy, PyAutoGUI
Databases & DevOps: PostgreSQL, Docker, Docker Compose
Developer Tools: Git & GitHub, VS Code, ESP32-S3
Soft Skills: Communication, Problem Solving, Team Collaboration, Research and Documentation

PROJECTS

AI-Powered Air Quality Monitoring System (ISRO Project) — **Nov 2025 – Present**
Python, LSTM, SciPy, C++, ESP32-S3
- **Role & Scope:** Collaborating as a System Engineer and Data Analyst Project Intern to develop a next-generation edge-computing AQMS for the Indian Space Research Organisation (ISRO).
- **ML Forecasting:** Co-developing a hybrid temporal forecasting pipeline with an LSTM network that ingests physics-validated outputs to predict real-time air quality and automate regulatory compliance tracking.
- **Physics-Based Modeling:** Assisting in the migration from a legacy PINN to a deterministic Chemical Transport Model (CTM), applying operator splitting for cleaner atmospheric data simulation.
- **Edge Systems:** Integrating an ESP32-S3 microcontroller with I2C/UART expansion bridges to ensure zero-loss, non-blocking sensor data feeds for the ML pipeline.

Automated Traffic Congestion MLOps System | Live Demo: API • Dashboard — **Jan 2026**
FastAPI, Pandas, Streamlit, Docker, Hugging Face Spaces
- Designed and deployed a real-time traffic congestion inference service using FastAPI, exposing a stateless REST API for low-latency ML predictions.
- Built a simulated live traffic pipeline by replaying historical Bengaluru traffic data in event-time order, enabling continuous inference and system-level testing without live sensor feeds.
- Developed a production-style monitoring dashboard with Streamlit to visualize traffic state, congestion level, severity score, and rolling predicted vs observed trends.
- Built and deployed separate inference and monitoring services on Hugging Face Spaces and Streamlit Cloud.

Product Recommendation App — **Nov 2025**
Python, Scikit-learn, Pandas
- Constructed content-based and collaborative filtering recommender systems, illustrating the effects of item similarity and user behavior on recommendation results.
- Applied TF-IDF with cosine similarity to generate product-to-product recommendations using textual metadata including descriptions and categories.
- Built a user-item matrix and k-nearest neighbors model to suggest items based on common user rating patterns and preference clustering.

AI-Powered NLP Query Engine — **Oct 2025**
Python, FastAPI, React, Docker
- Created a Text-to-SQL pipeline utilizing a Large Language Model (Gemini) with a dynamic schema discovery module.
- Built a full-stack application with hybrid search capabilities, incorporating semantic vector search. Containerized the complete application stack (Python, React, PostgreSQL) using Docker and DockerCompose.

CERTIFICATIONS

Artificial Intelligence: Knowledge Representation and Reasoning	NPTEL
Gen AI Powered Data Analytics Job Simulation	TATA iQ, Forage
Computer Graphics	NPTEL

EDUCATION

Cambridge Institute of Technology — **2023 – 2027**
Bachelor of Engineering (AI & ML) — *CGPA: 8.97 / 10*